Hardyal Singh Deora

Agentic AI Architect · Principal Software Engineer
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Open to remote roles from India, or hybrid anywhere in India

SUMMARY

Software architect with 15 years in backend and platform engineering, now designing and running LLM agent systems in production: tool use through Model Context Protocol (MCP), unattended scheduled agents, and guardrails and cost control for AI coding agents.

Brings the platform discipline agents need to be trusted: service boundaries, idempotency, audit trails, rollback plans and specifications other engineers build from.

Owns the architecture of a 13-microservice commercial platform on Amazon Web Services (AWS) with TypeScript, Python and PostgreSQL, and leads a squad while guiding about 10 engineers.

CORE SKILLS

Agentic AI: LLM agents, tool use / function calling, Model Context Protocol (MCP) servers and clients (stdio, Streamable HTTP), scheduled agent orchestration, human-in-the-loop review, guardrails, token and cost control, AI-assisted development (Claude Code)

LLM engineering: Retrieval-augmented generation (BM25 + Gemini embeddings, reciprocal rank fusion, cited sources), retrieval evaluation (hit@1 / hit@5 in CI), JSON-schema structured output (Zod, Gemini responseSchema), multi-provider integration (Anthropic Claude SDK, Google Gemini, local models via Ollama), context budgeting, image generation (Gemini image models)

Real-time AI (contributed): Voice agents on LiveKit/WebRTC, OpenAI and Gemini realtime models, speech-to-text build-vs-buy evaluation (managed APIs vs self-hosted Whisper)

Architecture: System and solution architecture, microservices, event-driven architecture, API gateway (JWT, rate limiting), backend-for-frontend, ports and adapters, idempotency, architecture decision records, build-vs-buy evaluation

Languages & Frameworks: TypeScript, Python, Node.js (Express, Hono), JavaScript, SQL, React, Angular, PHP (Laravel)

Data & Cloud: PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, ClickHouse, zero-downtime migration; AWS (S3, SQS, RDS, CloudFront), Azure, Docker, CI/CD

Leadership: Technical roadmaps, phased delivery, risk and rollback planning, stakeholder communication, mentoring senior engineers, Agile (SAFe, Scrum)

PROFESSIONAL EXPERIENCE

Principal Software Engineer
Educational Initiatives — Ahmedabad, India
07/2024 – Present

- Built and run an unattended LLM agent pipeline (TypeScript, Node.js) that reviews open pull requests every 30 minutes, calls tools through Model Context Protocol (MCP) and publishes a triaged report per repository.
- Built claude-toolkit, ten dependency-free CLI tools that add guardrails and cut token cost for Claude Code agent sessions.
- Contributed to a LiveKit/WebRTC voice-agent pipeline (OpenAI and Gemini realtime models, per-session transcription, lifecycle webhooks) and its speech-to-text build-vs-buy decision, costing managed APIs against self-hosted Whisper under data-protection constraints.
- Own end-to-end architecture for a B2B commercial platform of 13 microservices behind one API gateway: service boundaries, data models, contract-first integrations, JWT authentication and rate limiting at the edge.
- Directed a live MySQL to PostgreSQL migration across 6 product lines with no maintenance window (about 30 tables consolidated into 4, zero customer-visible incidents), with idempotency, retries, monitoring and rollback specified in a ~25,000-word document other engineers executed.

- Translated sales and finance requirements into a finance-auditable discount engine with a four-tier approval ladder, and payment mapping that cut reconciliation lag.
- Lead a squad of 4 engineers and guide about 10 across teams; set review, testing and coding standards and coach engineers on system design.

Lead Software Engineer
Educational Initiatives — Ahmedabad, India
04/2020 – 07/2024

- Designed and led a strangler migration of a live PHP/MySQL application to Node.js microservices on PostgreSQL with an Angular UI: 70 pages across 12 modules in 3 months with 2 developers, using a shared PHP-session + JWT login and per-module MySQL write flags so every product on the shared database kept working; no user-visible interruption, legacy app retired.
- Designed the order and subscription data model the platform still runs on, and a per-product order-creation REST API with OpenAPI (Swagger) documentation.
- Led sprint planning, backlog grooming and stakeholder communication across about 30 recorded sprints; grew the team from 2 full-time and 2 contract engineers with 30- and 90-day onboarding plans.
- Migrated a legacy product's full user base from PHP 5.3 to PHP 7.3, reworked CI/CD on Bitbucket Pipelines and shipped a versioned partner CRM integration (API v2).

Senior Software Engineer
Educational Initiatives — Ahmedabad, India
06/2016 – 04/2020

- Built sales-funnel and reporting dashboards, an internal tax module that removed manual finance steps, and diagnostic tooling for regional-language (Hindi) proficiency assessment.

Earlier roles
Analytics Solution, Ilogix Infotech, Seven Sleek, Softyoug Solution — Ahmedabad and Vapi, India
10/2011 – 06/2016

- Senior PHP Developer, PHP Developer and Front End Developer: Laravel and CodeIgniter platforms, REST APIs for mobile clients, and SEO-optimised web front ends.

PROJECTS

claude-toolkit — guardrail and token-cost tools for Claude Code agent sessions. github.com/hardyalsinghdeora/claude-toolkit

case-studies — public architecture decision record of the live migration. github.com/hardyalsinghdeora/case-studies

Multi-provider AI code reviewer (built, not yet launched) — Chrome extension reviewing Bitbucket pull requests through Anthropic Claude, Google Gemini or a local Ollama model behind one interface; schema-validated findings anchored to diff lines; noise-filtered, budgeted diffs. TypeScript, React.

Image-generation MCP server — MCP server giving Claude image generation and editing via Gemini image models; one tool module behind stdio and Streamable HTTP transports; keychain-held API key; container-ready for Cloud Run. Node.js.

Portfolio agent with hybrid RAG — Gemini function-calling agent on Cloudflare Workers that retrieves from a chunked career profile (BM25 + embeddings, reciprocal rank fusion) and cites its sources; retrieval eval set included. github.com/hardyalsinghdeora/portfolio

Ops Copilot (learning project) — LangGraph multi-agent RAG over MCP servers, LLM-as-judge review, human approval via interrupt(). github.com/hardyalsinghdeora/ops-copilot

EDUCATION

B.Tech, Electronics and Communications Engineering, Rajasthan Technical University (RTU), Kota — 2011

CERTIFICATIONS

Certified SAFe 5 Agilist, Scaled Agile — 2022

Google Project Management Certificate, Google / Coursera — 2022